YERUSLAN CHIHACHYOV

Decision Architect for High-Risk Operations

30+ Years in Marine, Offshore, FSO/FPSO and Production Operations

Operational Risk | Decision Integrity | Control Room Operations | Critical Controls

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Professional Profile

Experienced operational practitioner with **30+ years across marine, offshore, FSO/FPSO and production environments**, including control-room and production responsibilities in offshore environments where changing conditions, incomplete information, operational constraints and time pressure required continuous **judgment** and coordination. Progressed from marine mechanical operations into offshore production and CCR responsibilities, and subsequently developed an independent body of work focused on the structural conditions that enable timely intervention, clear decision ownership, effective controls and controlled escalation.

Founder of the **AVCS decision-integrity framework**, developed from accumulated operational observation and independent professional research.

Operational Responsibility & Experience

- **Control-room and production operations** – operational responsibility within FSO/FPSO production and CCR environments.
- **Machinery and marine systems** – operation, monitoring and maintenance of main and auxiliary machinery across multiple vessel types.
- **Abnormal and changing operating conditions** – response to alarms, deviations and abnormal situations under time pressure.
- **Operational coordination** – communication and coordination between control-room, field and technical functions.
- **Intervention and escalation** – participation in operational decisions during abnormal and emergency conditions.
- **Production continuity under constraints** – support of safe and stable production within real operating conditions.
- **Cross-functional interaction** – interaction between operational, technical, engineering and safety functions.

Professional Trajectory

MOTORMAN → SHIP MECHANICAL TECHNICIAN → OFFSHORE OPERATIONS → CCR / PRODUCTION OPERATOR → DECISION ARCHITECT

Core Competencies

Operations	Offshore FSO/FPSO Operations • Production Operations • Cargo & Ballast Operations • Control Room Operations • Marine Machinery & Systems • Operational Reliability
Operational Risk & Integrity	Critical Control Management • Barrier Management • Process Safety • Operational Drift • Procedure vs. Actual Practice • Root Cause Analysis • Incident Investigation
Decision Architecture	Decision Ownership • Authority Boundaries • Escalation Architecture • Protected Intervention • Override Visibility • Decision Traceability
Technical & Reliability	Predictive Maintenance • Condition Monitoring • Vibration Analysis • Thermal Monitoring • Rotating Equipment • Diesel Generators, Pumps, Compressors • Valve Repair
Coordination & Communication	Cross-Functional Operational Coordination • Communication Under Pressure • Control-Room Decision Support • Operational Problem Analysis • Field-to-Management Translation

Professional Experience

Founder & Decision Architect **2016 – Present**
Independent Research & Professional Practice – AVCS

- Developed **AVCS (Adaptive Vector Control System)** – a proprietary framework for structural integrity of operational decision systems, based on accumulated operational observation.
- **Analyze** how operational information, warnings, deviations and changing conditions are converted into **timely, authorized and reviewable action**.
- Examine the effectiveness of controls and procedures under abnormal, uncertain and time-critical conditions.
- **Analyze** decision ownership, authority boundaries, escalation pathways and intervention mechanisms.
- Examine gaps between documented procedures and actual operational **behavior**.
- Develop structural diagnostics for operational drift, unclear authority and weakened intervention pathways.
- Conduct case-based analysis of major high-risk operational failures and their decision environments.

Selected analytical areas: Deepwater Horizon | Piper Alpha | El Faro | Sewol Ferry | Cidade de Sao Mateus | Three Mile Island

FSO Cargo Operator **Aug 2009 – Aug 2016**
Petronas Carigali (Turkmenistan) Sdn Bhd – Caspian Sea, Diyarbekir Field

- Operated within the Permit-to-Work (PTW) system, following rules, regulations and work procedures applicable to cargo operations.
- Monitored, controlled and logged imported and produced oil; controlled and executed export oil discharge as instructed by CSUP.
- Monitored and executed ballast operations as instructed by CSUP.
- Monitored and controlled Fiscal Metering Unit (custody transfer) operations.
- Monitored and controlled unit trim, draft and hull stresses.
- Monitored and executed bulk chemicals loading and distribution operations.
- Monitored and recorded all safety equipment alarms received in CCR.
- Maintained radio communication with Process Deck, Export Tanker, Helicopter and supply boats during cargo and supply operations.

- Controlled supply operations and boat traffic in 500 m Zone.
- Maintained CCR Logbook and storage tank status records.
- Member of Emergency Response Team.
- Responded to critical equipment failures – cargo valve failures in cargo and ballast tanks, cargo pump failures, ballast pump malfunctions – assessing risk, coordinating with mechanical team, and maintaining safe cargo/ballast operations.
- Applied marine engineering background to detect early signs of equipment degradation – temperature changes in pump bearings, abnormal vibration in cargo and ballast pump casings – identifying potential failures well before breakdown, and coordinating preventive intervention with the mechanical team.
- Operated within FSO environments where H2S exposure was a recognized operational risk, adhering to strict H2S safety protocols and participating in regular emergency drills.
- **Recognized by Petronas Carigali management** as a competent and qualified specialist with no disciplinary cases, demonstrating leadership during solution of difficult operational tasks.

Operational strengths: decision-making under pressure • control-room situational awareness • recognition of developing operational deviations • practical understanding of procedure vs. operational reality

Pumpman — Apr 2006 – Jul 2009

Single Buoy Moorings Inc (now SBM Offshore) – Caspian Sea

- Supported CCR Operator in configuring cargo and ballast systems during loading and offloading operations.
- Monitored cargo operations on deck during offloading, taking periodic cargo measurements and reporting to CCR.
- Participated in repair of cargo and ballast pumps together with the engine room team.
- Replaced failed valves in cargo and ballast tanks; repaired cargo and ballast valves.
- Participated in cargo tank cleaning operations and crude oil sampling.
- Followed PTW system, rules, regulations and work procedures applicable to his responsibilities.
- Maintained radio communication with Process Deck, Export Tanker, Helicopter and supply boats as applicable.
- Member of Emergency Response Team.
- **Recognized by SBM Offshore management** for hard work and efforts that culminated in smooth and safe Production, Storage and Offloading Operations under difficult conditions on board MOPU and FSO.

Progressive Marine Engineering Roles — Sep 1993 – Apr 2006

Turkmen Maritime and River Lines (Turkmen Shipping)

- Progressed from motorman on fishing seiners and port tugs, through 4th, 3rd and 2nd engineer on dry cargo vessels and ore carriers, to **Senior Engineer on supply tugs and bunker vessels**.
- As Senior Engineer on supply tug, held full responsibility for the operation and maintenance of the vessel's engine room and machinery, working closely with the Captain.
- Repaired and maintained ship equipment including diesel generators, pumps, compressors, fuel separators, boiler equipment and valves.
- Led ship engineering teams; prepared technical reports and managed spare parts procurement.
- Responded to equipment failures under operational constraints, applying hands-on diagnostic and repair skills.

This period established the practical engineering foundation that later developed into offshore production

and control-room operations.

Operational Problem of Focus

How organizations maintain control when operating conditions change – particularly when warnings, procedures, authority, critical controls and operational reality begin to diverge.

AVCS – Independent Research & Professional Practice

Adaptive Vector Control System **2016 – Present**

Developed from accumulated experience in marine, offshore, FSO and control-room operations, AVCS formalizes recurring structural problems observed in high-risk operational environments.

AVCS addresses:

- Intervention – whether it is structurally protected and actionable
- Decision ownership – whether it is singular and real-time
- Escalation – whether it is mandatory or interpretive
- Critical controls – whether they remain effective under actual conditions
- Procedure vs. reality – where the two diverge and why
- Operational drift – normalization of deviations over time
- Evidence vs. status – whether claimed conditions can be independently verified
- Authority boundaries – where recommendation ends and authorization begins

AVCS does not replace engineering judgment, operational expertise, HSE systems, risk management or regulatory authority. Its purpose is to examine whether existing information, controls and procedures can be converted into timely, authorized and reviewable action when operating conditions change.

Research & Professional Work

Independent research into:

- Operational decision-making under pressure
- Structural integrity of operational decision systems
- Operational drift and normalization of deviations
- Authority and intervention architecture
- Procedure-versus-reality divergence
- Pre-incident vulnerability
- **Human decision-making under operational pressure**

Joint working research: Wojciech Z. Kaleta, PhD (Decision Engineering) – examining the relationship between decision epistemology and structural decision integrity. Joint Working Research Note, Version 1.1 (August 2026).

Education & Maritime Certificates

Vocational School No. 2 (SPTU No. 2) **1987 – 1990**
Qualification: Motorman 1st Class / Motorist 2nd Class
Secondary Specialized Technical Education

Turkmenbashi Liceum (Marine Lyceum No. 12) **1993 – 1995**
Diploma – Marine Engineering Technician
Qualification: Ship Mechanical Technician – Code 1620

Operation of Ship Power Plants
Technical Secondary Specialized Education

Certificates:

- **Pumpman Certificate** – Training Center "Novoship Training", Novorossiysk (2006) – STCW-78, Unlimited validity
- **GMDSS General Operator's Certificate** – PUSEM, Tuzla (Istanbul), 12/2010 – Present

Languages

Russian – Native | **English** – B2 (Upper Intermediate) | **Spanish** – B2 (Upper Intermediate) | **Turkmen** – B1 (Intermediate)

Professional Position

Decision Architect for High-Risk Operations

My professional focus sits at the intersection of:

Operational Reality → Reliability → Decision Integrity → Human Authority

I bring a field-based understanding of marine and offshore operations together with an independently developed framework for examining how decision systems behave under pressure.

Machines compute. Systems structure reality. Humans remain sovereign.